Hints: Number theory – 2

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. If r is a root of $x^3 - 3x + 1$, find another root that is a polynomial in r. Try the substitution $x = t + t^{-1}$.

Hint 2. Check the identity $(r^2-2)^3 - 3(r^2-2) + 1 = (r^3 - 3r - 1)(r^3 - 3r + 1)$. So $r^2 - 2$ is again a root.

Hint 3. Uniqueness forces $r^2 - 2 = r$, i.e. $r \in \{2, -1\}$, and then $p \mid 3$. Conversely, mod 3 the polynomial is $(x+1)^3$, so $a = 2$ is the only root. For $p = 2$ there are no roots. Answer: $p = 3$.

Problem 2

Note. As printed ("greater than $(p-3)/4$") the claim fails for $p = 7$: there the count is 1. The original Putnam statement has $p/4 - 1$. The hints prove that.

Hint 1. Rewrite the condition in terms of $j = I(k)$: since $jk \equiv 1$, $I(k+1) \equiv \frac{j}{j+1} = 1 - \frac{1}{j+1}$.

Hint 2. Put $m = j + 1$. The condition becomes $p + 1 - I(m) < m - 1$. So the count D equals $\#\{m \in [2, p-1] : m + I(m) > p + 2\}$.

Hint 3. The involution $m \mapsto p - m$ sends $m + I(m) = S$ to $2p - S$. Values $S \in \{p-1, p, p+1\}$ occur at most 6 times ($m^2 \equiv -1$, $m^2 \mp m + 1 \equiv 0$), and $S = p \pm 2$ never occurs. Hence $2D \geqslant p - 1 - 6$, which beats $p/4 - 1$ for $p > 10$. Check $p = 5, 7, 11$ directly.

Problem 3

Hint 1. For $N = 2^k$ put $D(n) = b_{n+N} - b_n$. From the recurrence, $D(n+1) = D(n)\,(1 + 2b_n + 2b_{n+N})$. So all $D(n)$ have the same 2-adic valuation as $D(0) = b_N$, and $b_{2N} - 2b_N = D(N) - D(0) = b_N(P-1)$ with $P = \prod_{n<N}(1 + 2b_n + 2b_{n+N})$.

Hint 2. Prove by induction on k: (i) $v_2(b_{2^k}) = k + 1$, and (ii) $Q_k = \prod_{n<2^k}(1 + 4b_n) \equiv 1 + 2^{k+1} \pmod{2^{k+2}}$. Base: $b_2 = 4$, $Q_1 = 5$.

Hint 3. By (i), $b_{n+N} \equiv b_n \pmod{2^{k+1}}$, so $P \equiv Q_k \pmod{2^{k+2}}$ and $v_2(P-1) = k+1$. This gives $v_2(b_{2N} - 2b_N) = 2k + 2$. Also $b_{2N} = b_N(P+1)$ with $P \equiv 1 \pmod 4$, which proves (i) for $k+1$. For (ii), $Q_{k+1} \equiv Q_k^2 \pmod{2^{k+3}}$ because $4D(n) \equiv 0 \pmod{2^{k+3}}$, and $(1 + 2^{k+1}u)^2 \equiv 1 + 2^{k+2}$.

Problem 4

Hint 1. Composite n: if $q^2 \mid n$, then q and $2q$ can only follow 1, which is impossible. If n is square-free with $s \geqslant 2$ prime factors, every even a_i must be followed by a quadratic residue mod n. Count: $\prod \frac{p_i+1}{2} < \frac{n-1}{2}$.

Hint 2. Prime $n = 2^k + 1$: let g be a primitive root. Then b can follow a exactly when $v_2(a) \leqslant v_2(\operatorname{ind} b)$, since the a-th powers are the $2^{v_2(a)}$-th powers. The two statistics $v_2(a)$ and $\mu(b) = v_2(\operatorname{ind} b)$ have the same distribution on $\{1, \ldots, 2^k\}$; for example, $a \mapsto g^a$ maps one level to the other.

Hint 3. Join each level $A_i = \{v_2 = i\}$ to $B_i = \{\mu = i\}$ by any bijection. This gives a disjoint union of cycles. Merge two cycles by swapping the successors of two vertices of the same level. A

cycle that cannot be merged contains all of A_λ and B_λ, where λ is its minimal level. But each B_i with $i \leqslant k-2$ contains an odd number (B_i is closed under $x \mapsto n-x$), and $B_{k-1} = \{2^k\} \to 1$. So that cycle contains an odd number, $\lambda = 0$, and odd vertices merge freely.

Problem 5

Note. The congruence $\equiv 1 \pmod{2^{k+2}}$ holds only for $k \geqslant 2$ ($F_1 = 5$).

Hint 1. For a Mersenne prime $q = 2^p - 1$, the order of 2 is p, since $2^j - 1 < q$ for $j < p$. For F_k see Problem 1 of the class work: $s = 2^{2^{k-2}}(2^{2^{k-1}} - 1)$ is a square root of 2 and has order 2^{k+2}.

Hint 2. Coprimality: $F_0 F_1 \cdots F_{m-1} = F_m - 2$, so a common prime factor of F_j and F_m divides 2. But the F_j are odd. Infinitely many pairwise coprime numbers > 1 give infinitely many primes.

Problem 6

Hint 1. The primitive roots are exactly g^k with $\gcd(k, p-1) = 1$, and there are $\varphi(p-1)$ of them.

Hint 2. Product: $k \mapsto p - 1 - k$ pairs these exponents without fixed points when $p > 3$. So the sum of the exponents is $\equiv 0 \pmod{p-1}$ and the product is 1.

Hint 3. Sum: use Möbius inversion on the condition $\gcd(k, p-1) = 1$. You get $\sum_{d \mid p-1} \mu(d) \sum_{x^{(p-1)/d}=1} x$. The inner sum, over all e-th roots of unity, is 0 unless $e = 1$, so only $d = p - 1$ survives.

Problem 7

Hint 1. Show: if a prime $q \nmid n$ divides $\Phi_n(a)$, then $\operatorname{ord}_q(a) = n$. Since $x^n - 1$ is separable mod q, a is not a root of any Φ_d with $d < n$, $d \mid n$. Hence $q \equiv 1 \pmod{n}$.

Hint 2. Take $a = n \cdot N!$. Then $\Phi_n(a) \equiv \pm 1$ modulo $n\,N!$, and $|\Phi_n(a)| > 1$, so it has a prime factor $> N$ not dividing n. Why the method stops there: the argument only ever produces primes whose order data force $q \equiv 1 \pmod{n}$. (A "Euclidean" polynomial proof exists only for classes a with $a^2 \equiv 1 \pmod{n}$; in general one needs Dirichlet's theorem.)

Problem 8

Hint 1. Small cases suggest primes. If m is prime and $k = m - 2n \geqslant 1$, then $k\binom{n}{k} = n\binom{n-1}{k-1}$, so $n \mid k\binom{n}{k}$, and $\gcd(n, k) = \gcd(n, m) = 1$.

Hint 2. Even $m = 2n > 2$: $\binom{n}{0} = 1$ is not divisible by n. Odd composite $m = pq$ with p the least prime factor, $q \geqslant p \geqslant 3$: take $n = (m-p)/2 \in [m/3, m/2]$, so $\binom{n}{m-2n} = \binom{n}{p}$.

Hint 3. $\binom{n}{p} = \frac{n}{p}\binom{n-1}{p-1}$, and $n - 1 \equiv -1 \pmod{p}$, so $\binom{n-1}{p-1} \equiv 1 \pmod{p}$ (Lucas). Hence $v_p\binom{n}{p} = v_p(n) - 1$ and $n \nmid \binom{n}{p}$. Answer: m prime.

Problem 9

Hint 1. Factor: $n^4 + n^2 + 1 = (n^2 - n + 1)(n^2 + n + 1) = g(n-1)g(n)$ with $g(t) = t^2 + t + 1$. Let $a(t)$ be the largest prime factor of $g(t)$. Note $g(t^2) = g(t)g(t-1)$.

Hint 2. $\gcd(g(t), g(t+1)) = 1$, so $a(t) \neq a(t+1)$. The required equality reads $\max(a(n-1), a(n)) = \max(a(n), a(n+1))$, which holds whenever $a(n)$ is a local maximum.

Hint 3. The sequence a is not eventually increasing, because $a(t^2) = \max(a(t), a(t-1))$. It is not eventually decreasing either: it would become constant, contradicting $a(t) \neq a(t+1)$. Hence it has infinitely many local maxima.

Problem 10

Hint 1. If $p \not\equiv 1 \pmod 5$, every residue is a fifth power: take $a = 0$. In general let $s_r = \#\{(a, b) : a^2 + b^5 \equiv r\}$, so that $\sum_r s_r = p^2$, and bound the number $N = \sum_r s_r^2$ of solutions of $a^2 + b^5 \equiv c^2 + d^5$.

Hint 2. For fixed (b, d), $a^2 - c^2 \equiv d^5 - b^5$ has $2p - 1$ solutions if the right side is 0 and $p - 1$ otherwise. The pairs with $b^5 \equiv d^5$ number at most $5p - 4$. Hence $N \leqslant p^3 + 4p^2 - 4p$.

Hint 3. Scaling $(a, b) \mapsto (\lambda^5 a, \lambda^2 b)$ maps S_r to $S_{\lambda^{10} r}$. So if some $s_r = 0$ with $r \neq 0$, at least $(p - 1)/10$ values of r have $s_r = 0$. Cauchy–Schwarz then gives $N \geqslant \frac{10p^4}{9p+1}$, which contradicts the bound for $p > 38$.